

New Design of RF Interface Circuits for PICC Complying with ISO/IEC14443-2 TypeB

Liang Lingyu, Zhao Nan, Yang Xingzi

Address: Institute of Microelectronics, Tsinghua University, Beijing, P.R.China, 100084

Email: LiangLingyu@tsinghua.org.cn

Abstract

This paper demonstrates a new set of RF circuits design proposal for PICC complying with ISO/IEC 14443-2 Type B, expatiates on the topology of typical modules and addresses its advantage over former design. The simulation data by Hspice, with the corresponding test of this chip is also given in the following thesis. The design has been implemented upon 0.8 μm CMOS technology successfully. The packaged chip can operate at 13.56MHz with data rate of 106 kbps, hence its feasibility is verified.

Keywords

RFID; PICC; High Performance; Full wave Rectifier; Modulation; Demodulation; Power-On-Off reset

Preface

RFID, Radio Frequency Identification, which can take contactless bidirectional communications for identification and data interchanging, which emerged in 1990s and get dramatically development throughout these years, fit itself as contactless IC card, among which PICC (Proximity IC Card) is the most popular one. Compared with some earlier card techniques, RFID is more favored for its long operating distance, high security, rapid processing speed and ideal convenience. Nowadays it has extended its way from mere object-identification to vast fields. This paper demonstrates the RF circuits design

proposal of high-load PICC that comply with ISO/IEC 14443-2 Type B. The design satisfies the performance request with neat circuits, and has been successfully implemented with 0.8 μm Wuxi CSMC-HJ CMOS technology.

Design Standard Choice

In ISO/IEC 14443-2, there are 2 types of RF interface about PICC operating at the frequency of 13.56MHz: Type A and Type B respectively. With regard to Type A, a patent maintained by Philips Semiconductor Co. Ltd, Type B is a worldwide open criterion. The signal of Type A from PCD (Proximity Coupled-Device) to PICC is modulated in ASK with modulation index 100%. Hence when receiving data from PCD, the PICC cannot process the data because of inadequate source driving, which makes the processing speed lowered. In Type B, the signal from PCD to PICC is modulated in 10% ASK, which make it possible for PICC to work continuously. Therefore the solution for Type B would simplify the design of circuits and reach high communication rate.

Circuits Design

In terms of circuits design, the contactless IC card system contains two main sections (Fig. 1): RF section and Digital Logic section. This paper focuses on the former, which is composed of several modules that resume the following functions apart: Power Recovery, Modulation, Demodulation, Power-On-Off

Reset, and Clock Generation & Division.

When the PICC moves into the electromagnetic field of the base station, the signal coupled by antenna coil can generate the power supply through Power Recovery module. As soon as the supply voltage reaches the certain value, the Power-On-Off Reset module gives out a low level signal, which changes the data processing unit to normal working status. The demodulator recovers the digital signal for data processing unit from the ASK signal, and the modulator modulates the digital signal from Digital Logic circuit for transmitting. Thus the communication between PICC and PCD can be established by simultaneously adjusting the duplexing of antenna.

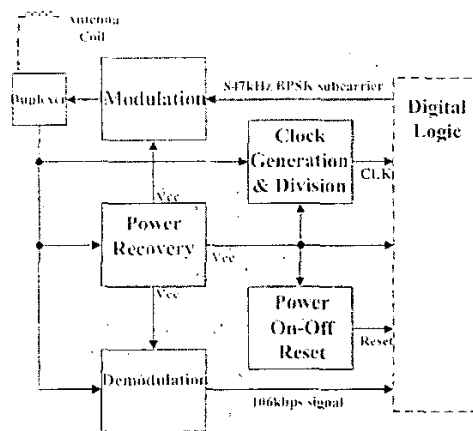


Fig. 1 Blockgram of RF for PICC

The individual module design is stated as follows.

1. Power Recovery module

As the source and the most important part of the RF front-end circuits, the Power Recovery module provides power supply for the whole chip by coupling the energy of the RF field transmitted by the PCD. The power supply is generated by the way of a full-wave rectifier that simplified as Fig. 2, which transforms the coupled AC to DC. Carefully choosing the LC parameter so as its resonance frequency is at 13.56MHz, we can get a high-efficiency DC power.

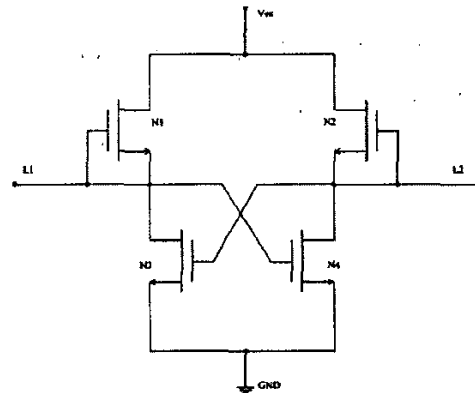


Fig. 2 Full-wave rectifier

The subsequent part is a high voltage restrictor in order to avoid excessively high source voltage, say, above 4V or more. This is accomplished in the circuit shown in Figure 3. Once the voltage across the Gate-Source voltage of PMOS Mp8 is high enough to exceed its threshold voltage, the MOS will be opened and the excessive energy will be discharged so as to lower the high voltage. The right-half circuit is to sample the supply voltage and control the gate voltage level of the power supply.

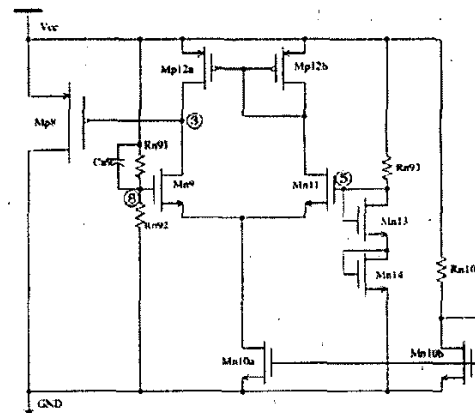


Fig. 3 High-voltage Restrictor

Compared with other circuits, this part would consume hardly any current when Mp8 is closed and conserves as much as energy to enhance the efficiency when the coupled signal is not strong.

Our simulation and test result shows that this circuit can drive a 300Ohm load with the voltage around 2.5V. (Fig. 4)

That's a great improvement and is enough to drive PICC with embedded low-power CPU.

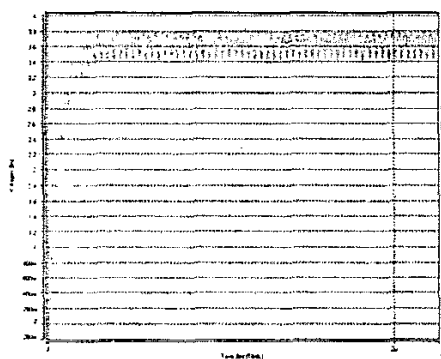


Fig. 4 Simulation waveform of PR

2. Demodulation module

The demodulator recovers the digital signals from the 10%ASK signal transmitted by PCD. In this section, a low pass filter and an extractor are needed to sense the fluctuation of the 106kbps envelope. How to sense the ripple of envelope and demodulate is the key design problem. It is a usual way to construct a bias circuit to generate a norm voltage and then compare the envelope with it to get the demodulated signal. But it's difficult to generate a bias voltage right between the peak and bottom of the envelope wave since the modulation depth is only 10%, or even less because of the noise. We present here a new design method. First let the envelope diverge into two transmission branches with different time delay. Therefore when the ripple comes, the two outputs of the circuit are separated. The Figure 5 shows the effect of the separation because of the different charge (or discharge) speed.

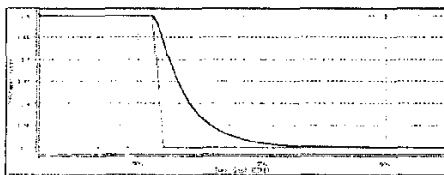


Fig. 5 Signal diverge of different delay path

Then with a simple comparator circuit we can get the demodulated signal. The Hspice simulation shows that this circuit can demodulate a 106kbps signal with modulation index above 1%, rather sufficient headroom for Type B PICC since its modulation depth is 10%. In other words, the demodulation is fulfilled.

In contrast to the former design method with bias circuit, this one is much easier to construct and can detect a much more noise-corrupted signal.

3. Modulation module

Traditional way to modulate is to change the transmitting parallel capacitor so that its coupled voltage would be affected. We design a new method by modulating the impedance paralleled across the antenna as the following Figure 6. And here TXData stands for the modulation signal from digital logic section.

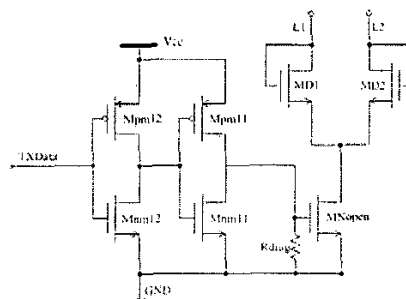


Fig. 6 Modulation Circuit

With this way we don't need to worry about the power supply efficiency, since the coupled resonant LC network hasn't been changed like the traditional method. Therefore the power efficiency is able to maintain a high driving capacity. This design idea is realized as the following figure. And the simulation in Fig. 7 confirms its feasibility.

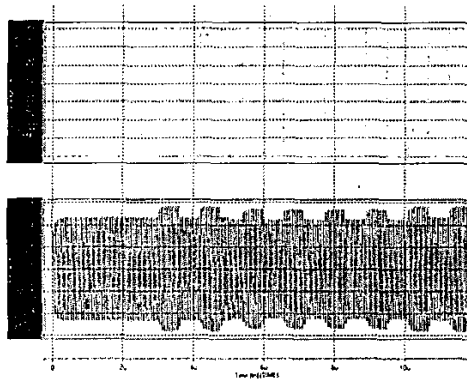


Fig. 7 Antenna waveform after modulation

4. Power On-Off Reset module

ISO/IEC 14443-2 requires the PICC to work continuously when electromagnetic intensity is between 1.5-7.5A/m rms, the giving out of the low-level signal must take these extreme conditions into account for keeping the working continuity of the circuits in PICC.

As stated before, a reset signal is required to notify the digital logic circuit when power supply voltage reaches a certain value at which the whole chip can work in gear. On the contrary, once the power is not sufficient to support the chip, this circuit should send another signal to close the digital logic. My design idea can be simplified as the following sketch. The capacitor C_0 is used as a delay regulator to control the reset signal generation delay time. With a Schmitt trigger, a power-on and a power-off signal are triggered under different supply voltage, achieving what we expected.

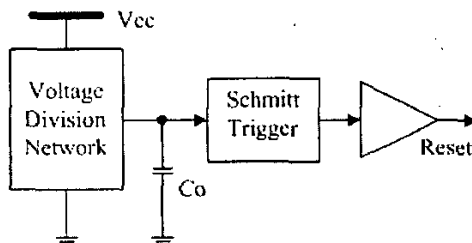


Fig. 8 Sketch of Power-On Reset Module

Compared to traditional circuits using a comparator, this circuit is fit for

PICC design because of its simplicity; for the PICC requires not a rigorously accurate power on-off value but low-power and succinct design.

5. Clock Generation & Division module

Without any local oscillator, the RF of PICC is required to generate from the carrier a frequency-divided digital signal through a reshaper and a frequency divider. In digital logic block, the working clock and the 847kHz subcarrier are generated from this signal. The Clock Generation circuit is shown in Figure 9 below, a fairly direct way in some sense.

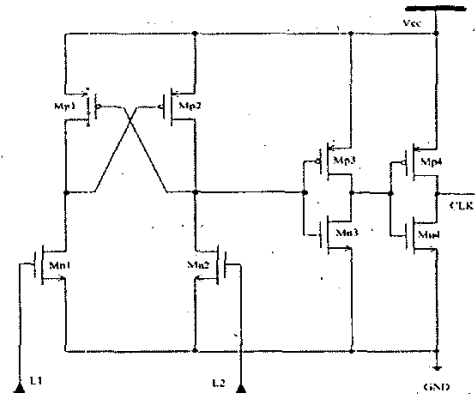


Fig. 9 Clock generation circuit

Performance Assessments

In the course of this design, each module and the overall system are simulated with the 0.8μm CMOS technology library, and the corresponding test of the chip verifies the feasibility of the design.

The Power Recovery module can drive 300Ω with 2.5V voltage, which means a driving DC above 8mA. The power on reset signal maintains low-level when the system turns into the steady state. When the modulation index of the signal received by antenna is above 1% with the data rate of 106kbps, the demodulator can recover the correct digital information. And the modulator superposes the BPSK signal on the carrier for transmitting.

Conclusion

The design addressed above has been successfully implemented upon the $0.8\mu\text{m}$ CMOS technology at Wuxi CSMC-HJ. The corresponding layout is demonstrated in the following **Figure 10**, which covers an area of about $1.12\text{mm} \times 0.87\text{mm}$.

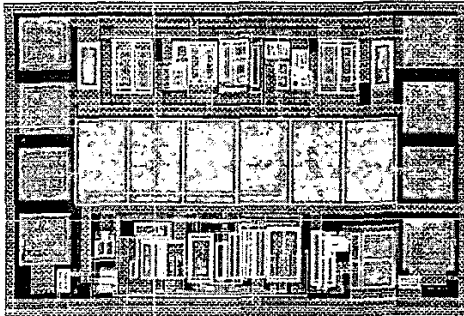


Fig. 10 Layout of RF front-end circuit

We have verified that this design has advantages over traditional design method. The packaged chip has been tested, and the result of the test is consistent with our former simulation. Our new RF front-end design for PICC satisfies the requirement of ISO/IEC 14443-2, and the test verifies that this chip operates well at 13.56MHz with the data rate of 106kbps.

References:

- [1] ISO/IEC14443-2, 2001, Identification cards Contactless integrated circuit(s) cards--Proximity cards--Part 2: Radio frequency power and signal interface
- [2] MicroID 13.56 MHz RFID System Design Guide, Microchip Technology Incorporated, 2001
- [3] U. Kaiser, W. Steinhagen, A Low Power Transponder IC for High Performance Identification Systems, IEEE Journal of Solid-State Circuits, Vol. JSSC 30, March 1995, pp.306-310
- [4] Lu Chao, Li Yong-ming, The RF Interface Circuits Design of Contactless IC Cards, ASICON 2001, Shanghai
- [5] Anantha P. Chandrakasan, Robert W. Brodersen, Minimizing Power

Consumption in CMOS Circuits, Reference of ECE of UC Berkeley

- [6] Alan Hastings, Roy Alan Hastings, The Art of Analog Layout, Prentice Hall, 1st edition, December 15, 2000